Mathematical Analysis — Midterm 2024, Row 1

Past exam · Răzvan

> **How to use.** The statement is on this page, exactly as in the official PDF (Class Materials → Past Exams → *midterm_2024_1.pdf*). Solve it on paper in **1h50min**, then turn to the solutions, which come after a red STOP page.

Midterm Test — Row 1

1. Prove that if the series $\sum_{n=1}^{\infty} x_n$ is convergent, then $\lim_{n\to\infty} x_n = 0$. 1p
2. Find the value of $\lim\limits_{n\to\infty}\left(\frac{1}{3n+1}+\frac{1}{3n+2}+\cdots+\frac{1}{4n}\right)$. 1p
3. Study the convergence of the following: (a) $\sum\limits_{n=2}^{\infty}\frac{1}{n\ln n}$. (b) $\int_0^{\pi}\frac{1}{\sin^2 x}\,dx$. (c) $\sum\limits_{n=1}^{\infty}\frac{1\cdot 4\cdot\ldots\cdot(3n-2)}{3^n\,n!}$. 3p
4. Find the Taylor series around 0 and its radius of convergence for each of the following: (a) $\ln(1-x)$ (1p); (b) $\sqrt{1+x}$ (1p).
5. Consider the power series $\sum\limits_{n=2}^{\infty}\frac{x^n}{n^2-n}$. (a) Find its radius of convergence and its convergence set. (1p) (b) Find its sum for any x in the convergence set. (1p)

Time 1h50min. Marking starts from 1p. Total 10p.

⛔ STOP
SOLUTIONS BELOW

De aici încep rezolvările

Rezolvă mai întâi subiectul singur, cronometrat (1h50min), și abia apoi dă pagina. Toate paginile cu rezolvări au o bandă roșie în stânga.

SOLUTIONS — Midterm 2024, Row 1

1. Necessary condition for convergence

Let $S_n = x_1 + \cdots + x_n$. Since the series converges, $S_n \to S \in \mathbb{R}$, and then also $S_{n-1} \to S$ (a subsequence with shifted index).

For $n \geq 2$, $x_n = S_n - S_{n-1}$, so by the algebra of limits

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} S_n - \lim_{n\to\infty} S_{n-1} = S - S = 0. \qquad \blacksquare$$

The converse is false: $\frac{1}{n} \to 0$ but $\sum \frac{1}{n}$ diverges. Contrapositive (used all the time): if $x_n \not\to 0$, the series diverges.

2. A limit as a Riemann sum

The sum has $4n - 3n = n$ terms; write it with $k = 1, \ldots, n$:

$$\sum_{k=1}^{n} \frac{1}{3n+k} = \frac{1}{n} \sum_{k=1}^{n} \frac{1}{3 + \frac{k}{n}}.$$

This is a Riemann sum of $f(x) = \frac{1}{3+x}$ on $[0,1]$, for the division $x_k = \frac{k}{n}$ (step $\frac{1}{n}$, intermediate points the right endpoints). f is continuous, hence integrable, so

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{3n+k} = \int_0^1 \frac{dx}{3+x} = \ln(3+x)\Big|_0^1 = \ln\frac{4}{3}.$$

$\Longrightarrow$ the limit is $\ln\frac{4}{3} \approx 0.2877$.

3. Convergence

(a) Positive, decreasing terms $\Longrightarrow$ Cauchy condensation test: $\sum a_n$ and $\sum 2^k a_{2^k}$ have the same nature.

$$2^k\, a_{2^k} = \frac{2^k}{2^k \ln 2^k} = \frac{1}{k \ln 2} \quad \Longrightarrow \quad \sum_k \frac{1}{k \ln 2} \text{ diverges (harmonic).}$$

$\Longrightarrow \sum \frac{1}{n \ln n}$ diverges.

Alternative (integral test): $\int_2^\infty \frac{dx}{x \ln x} = \ln(\ln x)\big|_2^\infty = \infty$.

(b) The integrand is unbounded at both ends ($\sin 0 = \sin \pi = 0$), so the integral is improper at 0 and at π. It converges only if both $\int_0^{\pi/2}$ and $\int_{\pi/2}^{\pi}$ converge. At 0: the antiderivative is $-\cot x$, and

$$\int_\varepsilon^{\pi/2} \frac{dx}{\sin^2 x} = -\cot x\Big|_\varepsilon^{\pi/2} = \cot\varepsilon \xrightarrow[\varepsilon\to 0^+]{} +\infty.$$

$\Longrightarrow$ the integral diverges.

Comparison view: $\sin x \leq x$, so $\frac{1}{\sin^2 x} \geq \frac{1}{x^2}$ on $(0, \frac{\pi}{2}]$, and $\int_0 \frac{dx}{x^2}$ diverges ($p = 2 \geq 1$).

(c) $a_n = \frac{1\cdot 4 \cdots (3n-2)}{3^n n!} > 0$. Ratio:

$$\frac{a_{n+1}}{a_n} = \frac{3n+1}{3(n+1)} \to 1 \quad \text{(inconclusive).}$$

Raabe–Duhamel test:

$$n\left(\frac{a_n}{a_{n+1}}-1\right)=n\cdot\frac{3n+3-(3n+1)}{3n+1}=\frac{2n}{3n+1}\to\frac{2}{3}<1.$$

$\Longrightarrow$ **the series diverges.**

4. Taylor series

(a) Start from the geometric series $\frac{1}{1-t}=\sum_{n\ge0}t^n$, $|t|<1$. Integrate term by term on $[0,x]$ (allowed inside the interval of convergence):

$$-\ln(1-x)=\int_0^x\frac{dt}{1-t}=\sum_{n\ge0}\frac{x^{n+1}}{n+1}\quad\Longrightarrow\quad\ln(1-x)=-\sum_{n=1}^{\infty}\frac{x^n}{n}.$$

Radius: $R=\lim\frac{|c_n|}{|c_{n+1}|}=\lim\frac{n+1}{n}=1$.

$\Longrightarrow\ln(1-x)=-\sum_{n\ge1}\frac{x^n}{n}$**,** $R=1$ **(equality on** $[-1,1)$**).**

(b) Binomial series with $\alpha=\frac12$: $(1+x)^\alpha=\sum_{n\ge0}\binom{\alpha}{n}x^n$, $\binom{\alpha}{n}=\frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}$. For $\alpha=\frac12$ and $n\ge2$:

$$\binom{1/2}{n}=\frac{\frac12\left(-\frac12\right)\left(-\frac32\right)\cdots\left(\frac{3-2n}{2}\right)}{n!}=(-1)^{n-1}\frac{1\cdot3\cdots(2n-3)}{2^n\,n!}.$$

$$\sqrt{1+x}=1+\frac{x}{2}-\frac{x^2}{8}+\frac{x^3}{16}-\frac{5x^4}{128}+\cdots=1+\frac{x}{2}+\sum_{n\ge2}(-1)^{n-1}\frac{1\cdot3\cdots(2n-3)}{2^n\,n!}x^n.$$

Radius: $\frac{|c_n|}{|c_{n+1}|}=\frac{2(n+1)}{2n-1}\to1$.

$\Longrightarrow R=1$**.**

5. Power series

(a) $c_n=\frac{1}{n(n-1)}$, $\frac{c_n}{c_{n+1}}=\frac{n+1}{n-1}\to1\Longrightarrow R=1$. Endpoints: at $x=\pm1$, $\left|\frac{(\pm1)^n}{n(n-1)}\right|=\frac{1}{n(n-1)}\le\frac{2}{n^2}$, so the series converges absolutely (comparison with $\sum\frac{1}{n^2}$).

$\Longrightarrow R=1$**, convergence set** $[-1,1]$**.**

(b) Split with partial fractions, $\frac{1}{n(n-1)}=\frac{1}{n-1}-\frac1n$. For $|x|<1$ both series below converge, so we may subtract them:

$$\sum_{n\ge2}\frac{x^n}{n-1}=x\sum_{m\ge1}\frac{x^m}{m}=-x\ln(1-x),\qquad\sum_{n\ge2}\frac{x^n}{n}=-\ln(1-x)-x.$$

$$S(x)=-x\ln(1-x)+\ln(1-x)+x=(1-x)\ln(1-x)+x,\qquad x\in[-1,1).$$

(At $x=-1$ the formula still holds, by Abel's theorem: the series converges there and S is continuous.) At $x=1$ the sum telescopes: $\sum_{n\ge2}\left(\frac{1}{n-1}-\frac1n\right)=1$; this also equals $\lim_{x\to1^-}\left[(1-x)\ln(1-x)+x\right]=1$.

$\Longrightarrow S(x)=(1-x)\ln(1-x)+x$ **for** $x\in[-1,1)$**, and** $S(1)=1$**.**

Results at a glance

Ex.	Answer	Tool
1	$x_n=S_n-S_{n-1}\to0$	partial sums
2	$\ln\frac43$	Riemann sum

3	(a) diverges; (b) diverges; (c) diverges	condensation; $-\cot x$; Raabe ($\frac{2}{3} < 1$)
4	$-\sum \frac{x^n}{n}$, $R = 1$; binomial series, $R = 1$	integrate geometric series; binomial series
5	$[-1, 1]$; $S = (1 - x)\ln(1 - x) + x$, $S(1) = 1$	partial fractions